

I

CRUSH INJURIES

TREATMENT IS TO AVOID THE MAJOR COMPLICATIONS OF HYPOVOLAEMIC SHOCK OR CRUSH SYNDROME CAUSING ARRHYTHMIA (HYPERKALAEMIA/HYPOCALCAEMIA), RHABDOMYOLYSIS, and ACUTE KIDNEY INJURY (AKI)/ACUTE RENAL FAILURE (ARF).

MAJOR HAEMORRHAGE CONTROL — AIRWAY — ASSESS CONSCIOUS LEVELS

CONSIDERATIONS OF EXTRICATION — IS CRITICAL CARE REQUIRED?

TIMEFRAME, OBJECT WEIGHT AND MUSCLE BULK ENTRAPPED

URGENCY OF EXTRICATION (BIG SICK OR LITTLE SICK?)

SERIOUS INJURIES THAT CANNOT BE MANAGED IN CONFINED AREA

PRESSURES ON THE HEAD OR TORSO REQUIRED TO CAUSE CRUSH SYNDROME ARE CONSIDERED UNSURVIVABLE – ASSESS FOR SIGNS OF LIFE AND R.O.L.E CRITERIA⁽¹⁾

APPLY TWO TOURNIQUETS SIDE BY SIDE IF:

- PROLONGED PHYSICAL ENTRAPMENT >2 HOURS⁽²⁾
- OBVIOUS MAJOR HAEMORRHAGE PRESENT
- LIMB IS ASSUMED TO HAVE MASSIVE INJURY FROM HEAVY WEIGHT IRRESPECTIVE OF TIMEFRAME
- IMMEDIATE EXTRICATION IS REQUIRED WITH NO TIME TO SUITABLY ASSESS

DEAD LIMB

- ◇ COLD
- ◇ NO SENSORY OR MOTOR FUNCTION
- ◇ PULSELESS
- ◇ TENSELY SWOLLEN
- ◇ SEVERE PALLOR OR CYANOSIS

CONSIDER PREHOSPITAL AMPUTATION IF TRAINED OR TRANSFER TO SURGICAL CAPABILITY URGENTLY **A**

ASSESSMENT

CONTINUOUS ECG MONITORING FOR ARRHYTHMIAS⁽³⁾
MONITOR FOR SIGNS OF SHOCK⁽⁴⁾

MEASURE POTASSIUM (3.5 – 5.2mmol/L) AND CREATINE KINASE (22 – 198 U/L)
URINARY CATHETER IS ADVISABLE TO MONITOR URINARY OUTPUT IF TRAINED TO DO SO

IMMEDIATE TREATMENT - GAIN IV/IO ACCESS

ADMINISTER A 1-2L BOLUS OF CRYSTALLOID FLUID BEFORE EXTRICATION OR RELEASE OF TOURNIQUETS
FLUID MAINTENANCE 1L PER HOUR TO MAINTAIN URINARY OUTPUT OF 1-2ML/KG/HR TO AVOID ARF/AKI
ADMINISTER SODIUM BICARBONATE (NaHCO₃) 8.4% 50ML EVERY 4th OR 5th BAG OF FLUID TO AVOID ARF/AKI
ONCE NORMOVOLAEMIA IS ACHIEVED, ALTERNATE BETWEEN CRYSTALLOID AND 5% GLUCOSE
CONSIDER TXA AND ANALGESIA WITH SPLINTING AND APPROPRIATE MEDICATIONS

SEVERE COMPLICATIONS OF CRUSH INJURY

RB

CRUSH SYNDROME

ARRHYTHMIA OR HYPERKALAEMIA

ANY HYPERACUTE T WAVES or
WIDENING QRS or PROLONGED QTc
or K⁺ >5.5mmol/L

AGGRESSIVE TREATMENT WITH:
- CALCIUM CHLORIDE (CaCl₂) 10% 10ml
(LARGE FLUSHES WITH 5% GLUCOSE)
- CONSIDER IV INSULIN 10 UNITS WITH
IV GLUCOSE 10% 500ml or 50% 50ml
- CONSIDER SALBUTAMOL 10-20mg
NEB
- CONSIDER NaHCO₃ (SEPARATE IV)

ACUTE RENAL FAILURE

PRESENTS WITH MYOGLOBINURIA
(DARK URINE) OR OLIGURIA
(<400ml OUTPUT)

AGGRESSIVE TREATMENT WITH:
- IV CRYSTALLOID FLUIDS 1-2ml/kg/hr
- NaHCO₃ 8.4% 50ml EVERY 4th 500ml
BAG OF FLUID
- WILL REQUIRE AT LEAST 12L/DAY AND
URINARY OUTPUT SHOULD BE 8L/DAY
- IF NORMOVOLAEMIC, ALTERNATE
BAGS OF FLUID WITH 5% GLUCOSE.

COMPARTMENT SYNDROME

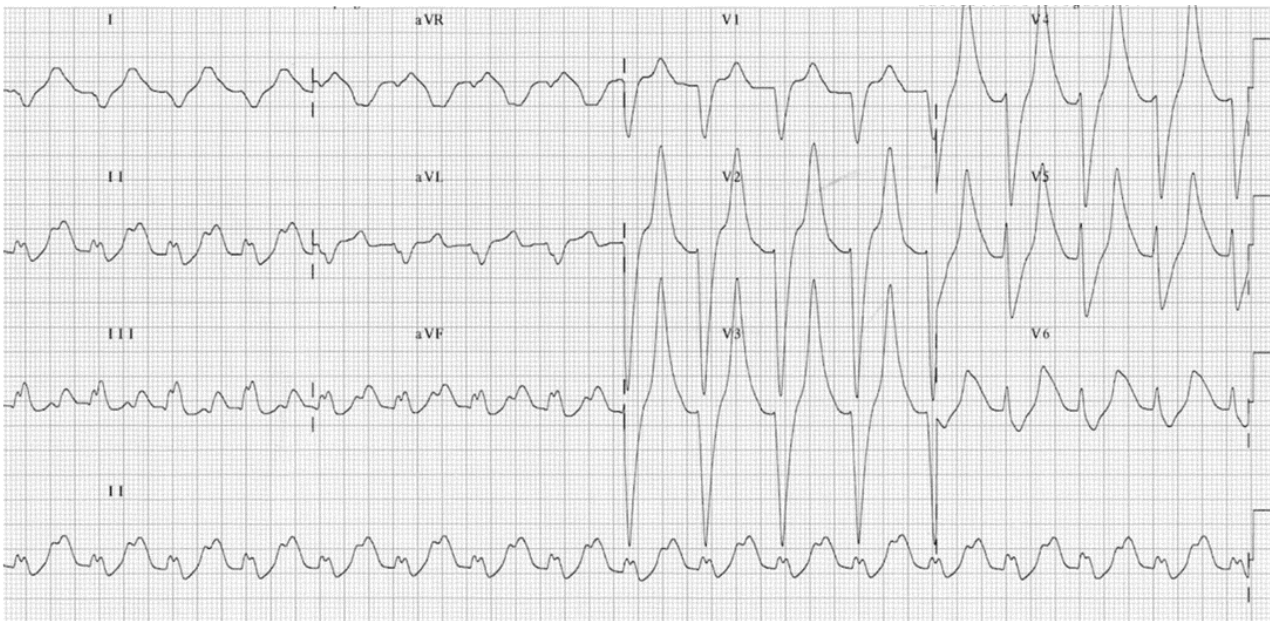
PAIN (SEVERE)
PARAESTHESIA
POIKILOthermia
PALLOR
PARALYSIS
PULSELESSNESS

ADMINISTER MANNITOL 20% 500ml

CONSIDER FASCIOTOMY IF TRAINED:
<6 HOURS ENTRAPPED
LIMB IS VIABLE (NOT DEAD LIMB)
NOT SYSTEMICALLY HYPOVOLAEMIC

POINTS TO NOTE

1. CRUSH OF THE HEAD OR TORSO SIGNIFICANT ENOUGH TO CAUSE CRUSH SYNDROME IS CONSIDERED TO BE FATAL DUE TO INHERENT INTERNAL ORGAN DAMAGE. RECOGNITION OF LIFE EXTINCT CRITERIA INCLUDES
 - ⇒ DECAPITATION
 - ⇒ HEMICORPECTOMY
 - ⇒ MASSIVE CRANIAL OR CEREBRAL DESTRUCTION
2. CRUSH SYNDROME IS A MORE SIGNIFICANT CONCERN THAN LIMB ISCHAEMIA IN THE SETTING OF PROLONGED CRUSH INJURY.
3. ARRHYTHMIAS CAN OCCUR DUE TO HYPOCALCAEMIA AND HYPERKALAEMIA. NOTE ANY PEAKED T WAVES (MILD), LOSS OF P WAVES (MODERATE) OR BROAD QRS COMPLEXES (SEVERE) THAT COULD SUGGEST HYPERKALAEMIA (SEE ECG PICTURE BELOW). TREAT AGGRESSIVELY AND BE READY TO PROVIDE ALS.
4. HYPOTENSION IN THE ACUTE PHASES OF CRUSH INJURIES SHOULD BE PRESUMED TO BE FROM HYPOVOLAEMIA UNTIL PROVEN OTHERWISE.
5. PERMISSIVE HYPOTENSION IS ONLY TO BE CONSIDERED IF THERE IS INCOMPRESSIBLE INTERNAL HAEMORRHAGE BUT OTHERWISE SHOULD NOT BE USED IN THE SETTING OF CRUSH INJURY AS IT CAN WORSEN ACUTE RENAL FAILURE.
6. URINARY OUTPUT IS THE BEST INDICATOR OF RENAL FUNCTION AND MEASUREMENT OF APPROPRIATE FLUID RESUSCITATION. AIM FOR 1-2ml/kg/hr FOR RENAL PROTECTION.
7. ANY DARK URINE INDICATES MYOGLOBINURIA AND MUST BE MANAGED WITH FLUID RESUSCITATION.
8. IN HEALTHY YOUNG ADULTS, UP TO 12L OF IV FLUID IN 24 HOURS MAY BE REQUIRED.
9. TO MAKE 5% GLUCOSE FROM 10% GLUCOSE, REMOVE 250ML OF 10% SOLUTION AND DILUTE WITH 250ML OF SODIUM CHLORIDE FOR 5% GLUCOSE IN 500ML.



Guideline written in accordance with SFM Clinical Standard Operating Procedure 012 'Crush Injury' (2022) and MERT Standard Operating Procedure 'Crush V3.0'

References

- Greaves, I., Porter, K. and Smith, J. E (2003) Consensus Statement on the Early Management of Crush Injury and Prevention of Crush Syndrome, *Journal of Royal Army Medical Corps*, 149(4), pp. 255-259.
- Schwartz, D. S., Weisner, Z. and Badar, J. (2015) Immediate Lower Extremity Tourniquet Application to Delay Onset of Reperfusion Injury after Prolonged Crush Injury, *Prehospital Emergency Care*, 19(4), 544-547, DOI: 10.3109/10903127.2015.1005264
- Sever, M. S. and Vanholder, R. (2013) Management of Crush Victims in Mass Disasters: Highlights from Recently Published Recommendations, *CJASN*, 8(2), pp. 328-335; DOI: 10.2215/CJN.07340712